

Operation & Maintenance Manual

Division 23 - Heating, Ventilating and Air Conditioning

PROJECT	Harborview Family Clinic - Tenant Fit-Out (fictional project)
PROJECT NUMBER	2026-014
OWNER	Harborview Health Partners (fictional)
GENERAL CONTRACTOR	Quellmoor Builders (fictional)
MECHANICAL SUBCONTRACTOR	Northgate Mechanical Services (fictional)
SUBSTANTIAL COMPLETION	September 18, 2026
VOLUME	1 of 1
PREPARED BY	Sample Document Services
DATE	October 7, 2026

16

DOCUMENTS

1

PENDING

8

SECTIONS

44

PAGES

Compiled from manufacturer and contractor documents as supplied. Documents have not been altered. Warranty terms are those stated in each warranty document.

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Harborview Family Clinic (fictional) - Division 23 HVAC O&M Manual

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SECTION

1

General Information

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NORTHGATE MECHANICAL SERVICES (FICTIONAL)

Harborview Family Clinic (fictional) · Project 2026-014

Project contacts

Who to call about the HVAC installation (sample)

Role	Company (fictional)	Contact (fictional)	Phone	Email
Owner	Harborview Health Partners	Pat Example	555-0101	pat@example.com
General contractor	Quellmoor Builders	Robin Sample	555-0102	robin@example.com
Mechanical subcontractor	Northgate Mechanical Services	Alex Placeholder	555-0103	alex@example.com
Controls	Lumenwise Controls (service)	Jordan Demo	555-0104	jordan@example.com
Test and balance	Balanced Air Testing	Casey Mock	555-0105	casey@example.com
Rooftop unit supplier	AeroCrest Climate Systems	Service desk	555-0106	service@example.com

All names, numbers and email addresses are fictional (555-01xx numbers and example.com addresses are reserved for examples).

NORTHGATE MECHANICAL SERVICES (FICTIONAL)

Harborview Family Clinic (fictional) · Project 2026-014

Warranty summary

Equipment warranties for Division 23 (sample)

Equipment	Tag	Manufacturer (fictional)	Warranty (sample)	Starts	Ends
Packaged rooftop unit	RTU-1	AeroCrest	1 yr parts; 5 yr heat exchanger	2026-09-18	2031-09-18
Inline exhaust fans	EF-1 to EF-3	Ventora	2 yr parts	2026-09-18	2028-09-18
VAV terminal units	VAV-1 to VAV-4	Tessel	1 yr parts	2026-09-18	2027-09-18
Thermostat	T-1	Lumenwise	1 yr parts	2026-09-18	2027-09-18
Installation workmanship	All	Northgate Mechanical	1 yr	2026-09-18	2027-09-18

Sample summary. The warranty documents in this binder state the actual (sample) terms.

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Packaged Rooftop Unit RTU-1 (23 74 00)

IN THIS SECTION		PAGE
2.1	Product data - AeroCrest RTU-480 series Unit as installed: RTU-480-060	7
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AeroCrest RTU-480 Series

Packaged rooftop unit, 3 to 10 nominal tons (sample)

Sample product description. A real datasheet describes the unit, its options and its certifications here. Every figure in this document is invented.

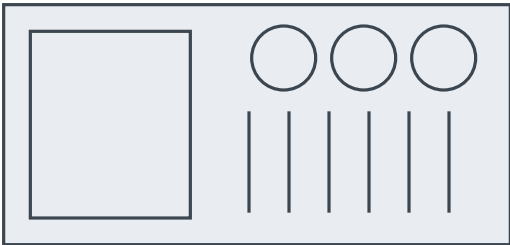


Illustration only (sample).

FEATURES

- Single-piece packaged design for rooftop curb mounting (sample text).
- Factory-installed economizer option (sample text).
- Service access panels on one side (sample text).

SPECIFICATIONS (SAMPLE VALUES)

Item	Value
Nominal capacity	5 tons (sample)
Heating	Gas, 120 MBH input (sample)
Electrical	208-230 V / 3 ph / 60 Hz (sample)
Filters	(4) 20 x 25 x 2 in (sample)
Operating weight	780 lb (sample)
Refrigerant	Sample refrigerant

PERFORMANCE AND ELECTRICAL DATA (SAMPLE VALUES)

Model	Cooling (sample)	Heating (sample)	Electrical (sample)
RTU-480-036	3.0 tons	80 MBH	208-230/3/60
RTU-480-060	5.0 tons	120 MBH	208-230/3/60
RTU-480-090	7.5 tons	180 MBH	460/3/60
RTU-480-120	10.0 tons	240 MBH	460/3/60

DIMENSIONS



All values on this page are invented for a sample. Not for design or ordering.

AEROCREST CLIMATE SYSTEMS

AeroCrest RTU-480 Series

Installation, Operation and Maintenance Manual

Manual AC-IOM-480 Rev B (sample)

SAMPLE MANUAL

Contents

1	Safety	3
2	Installation	4
3	Start-up and operation	5
4	Maintenance	6

SAMPLE MANUAL

1 Safety

Sample text only. A real manual lists the manufacturer's safety instructions here. Follow the real manufacturer's instructions for real equipment.

SAMPLE MANUAL

2 Installation

Sample text only. A real manual explains rigging, mounting and connections here.

Sample text only. Wiring diagrams normally follow.

SAMPLE MANUAL

3 Start-up and operation

Sample text only. A real manual gives the start-up checklist here.

SAMPLE MANUAL

4 Maintenance

Sample text only. The table below shows the kind of schedule a real manual gives; the intervals are invented.

Task (sample)	Interval	Notes
Inspect filters	Sample	Sample
Check belts and bearings	Sample	Sample
Clean coils	Sample	Sample

AEROCREST CLIMATE SYSTEMS

LIMITED WARRANTY CERTIFICATE

Sample document. Terms below are invented and have no legal effect.

Project	Harborview Family Clinic (fictional)
Equipment	Packaged rooftop unit RTU-1
Model	RTU-480-060 (sample)
Serial number	SAMPLE-0000-0001
Warranty start	September 18, 2026 (substantial completion)

Coverage (sample terms)

- Parts: 1 year from start (sample term).
- Heat exchanger: 5 years from start (sample term).
- Labor is not included (sample term).

Authorized representative (fictional)

Date

SECTION

3

Exhaust Fans EF-1 to EF-3 (23 34 23)

IN THIS SECTION		PAGE
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Ventora EF-12 Inline Exhaust Fan

Inline centrifugal exhaust fan (sample)

Sample product description printed on A4 paper, as many European manufacturers do. All values are invented.

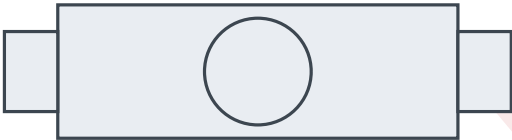


Illustration only (sample).

FEATURES

- Inline mounting in round duct (sample text).
- Speed controller option (sample text).

SPECIFICATIONS (SAMPLE VALUES)

Item	Value
Airflow	420 l/s (sample)
Static pressure	250 Pa (sample)
Motor	0.37 kW, 230 V / 1 ph (sample)
Duct connection	315 mm (sample)

VENTORA AIR MOVEMENT

Ventora EF-12 Inline Exhaust Fan

Installation, Operation and Maintenance Manual

Manual VT-EF12-IOM (sample)

SAMPLE MANUAL

Contents

1	Safety	3
2	Installation	4
3	Maintenance	5

SAMPLE MANUAL

1 Safety

Sample text only. A real manual lists the manufacturer's safety instructions here. Follow the real manufacturer's instructions for real equipment.

SAMPLE MANUAL

2 Installation

Sample text only. A real manual explains rigging, mounting and connections here.

Sample text only. Wiring diagrams normally follow.

SAMPLE MANUAL

3 Maintenance

Sample text only. The table below shows the kind of schedule a real manual gives; the intervals are invented.

Task (sample)	Interval	Notes
Inspect filters	Sample	Sample
Check belts and bearings	Sample	Sample
Clean coils	Sample	Sample

VENTORA AIR MOVEMENT

LIMITED WARRANTY CERTIFICATE

Sample document. Terms below are invented and have no legal effect.

Project	Harborview Family Clinic (fictional)
Equipment	Exhaust fans EF-1 to EF-3
Model	EF-12 (sample)
Serial numbers	SAMPLE-EF-0001 to 0003
Warranty start	September 18, 2026

Coverage (sample terms)

- Parts: 2 years from start (sample term).

Authorized representative (fictional)	Date
---------------------------------------	------

SECTION

4

VAV Terminal Units (23 36 00)

IN THIS SECTION

PAGE

4.1	Product data - Tessel VAV-S	25
4.2	Installation and maintenance manual	27

Tessel VAV-S Single-Duct Terminal

Pressure-independent VAV terminal unit (sample)

Sample product description for a single-duct terminal unit. All values are invented.

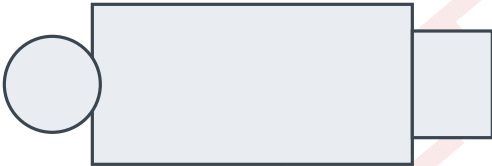


Illustration only (sample).

FEATURES

- Pressure-independent airflow control (sample text).
- Optional hot-water reheat coil (sample text).

SPECIFICATIONS (SAMPLE VALUES)

Item	Value
Inlet sizes	6, 8, 10, 12 in (sample)
Airflow range	100-2000 cfm (sample)
Casing	22 ga galvanized steel (sample)
Controls	Factory-mounted actuator (sample)

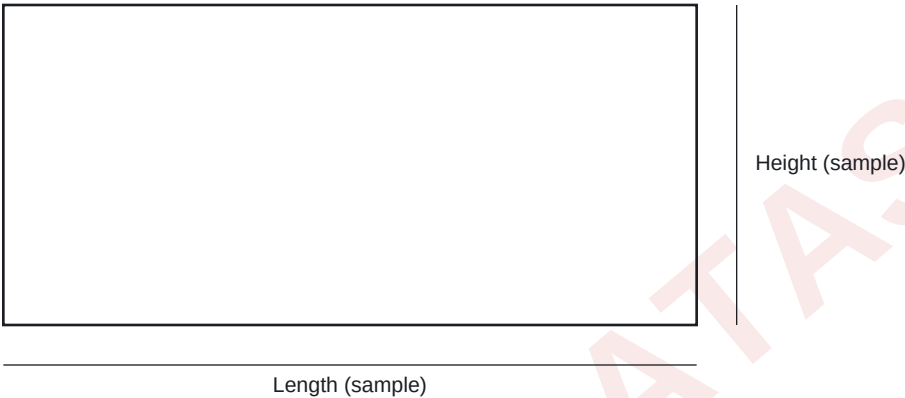
TESSEL AIRFLOW PRODUCTS

PRODUCT DATA

PERFORMANCE AND ELECTRICAL DATA (SAMPLE VALUES)

Inlet size	Min cfm (sample)	Max cfm (sample)	Notes
6 in	100	500	Sample
8 in	180	900	Sample
10 in	280	1400	Sample
12 in	400	2000	Sample

DIMENSIONS



All values on this page are invented for a sample. Not for design or ordering.

TESSEL AIRFLOW PRODUCTS

Tessel VAV-S Single-Duct Terminal

Installation, Operation and Maintenance Manual

Manual TA-VAVS-IOM (sample)

SAMPLE MANUAL

Contents

1	Installation	3
2	Start-up and operation	4
3	Maintenance	5

SAMPLE MANUAL

1 Installation

Sample text only. A real manual explains rigging, mounting and connections here.

Sample text only. Wiring diagrams normally follow.

SAMPLE MANUAL

2 Start-up and operation

Sample text only. A real manual gives the start-up checklist here.

SAMPLE MANUAL

3 Maintenance

Sample text only. The table below shows the kind of schedule a real manual gives; the intervals are invented.

Task (sample)	Interval	Notes
Inspect filters	Sample	Sample
Check belts and bearings	Sample	Sample
Clean coils	Sample	Sample

SECTION

5

HVAC Controls (23 09 00)

IN THIS SECTION

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5.2	Sequence of operations	34
	As installed	

Lumenwise T-200 Thermostat

Programmable thermostat for rooftop units (sample)

Sample product description for a wall thermostat. All values are invented.

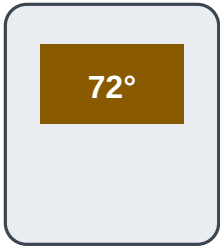


Illustration only (sample).

FEATURES

- 7-day schedule (sample text).
- Occupancy override button (sample text).

SPECIFICATIONS (SAMPLE VALUES)

Item	Value
Power	24 VAC (sample)
Stages	2 heat / 2 cool (sample)
Display	Backlit (sample)

NORTHGATE MECHANICAL SERVICES (FICTIONAL)

Sequence of operations - RTU-1 and VAV system

As installed (sample text)

Occupied mode

Sample text only. A real sequence describes how the unit runs when the building is occupied.

Unoccupied mode

Sample text only.

Economizer

Sample text only.

Alarms

Sample text only.

NORTHGATE MECHANICAL SERVICES (FICTIONAL)

End of sample sequence of operations.

SAMPLE DOCUMENT

SECTION

6

Maintenance

IN THIS SECTION

PAGE

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	Handed over to the owner	

NORTHGATE MECHANICAL SERVICES (FICTIONAL)

Harborview Family Clinic (fictional) · Project 2026-014

Preventive maintenance schedule

Summary from the manufacturers' manuals (sample; intervals invented)

Equipment	Task	Frequency (sample)	Reference
RTU-1	Replace filters	Sample	AeroCrest manual, ch. 4
RTU-1	Inspect belts, clean coils	Sample	AeroCrest manual, ch. 4
EF-1 to EF-3	Check bearings and wheel	Sample	Ventora manual, ch. 3
VAV-1 to VAV-4	Check actuator and airflow	Sample	Tessel manual, ch. 3
T-1	Check schedule and battery	Sample	Lumenwise datasheet

Sample document. Real maintenance follows the manufacturers' instructions.

Spare parts and attic stock

Items handed over to the owner (sample)

Item	For	Quantity	Location (sample)
Filter set 20 x 25 x 2	RTU-1	2 sets	Storage room
Fan belt (sample part no.)	RTU-1	1	Storage room
Thermostat cover key	T-1	2	Office

SECTION

7

Testing, Adjusting and Balancing (23 05 93)

IN THIS SECTION

PAGE

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	Includes a rotated landscape page	
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	PENDING — NOT YET RECEIVED To be provided by the balancing contractor	

Preliminary air balance readings

Taken before final balancing (sample values)

Item	Design (sample)	Measured (sample)	Notes
RTU-1 supply	2000 cfm	1940 cfm	Sample
EF-1	300 cfm	290 cfm	Sample
EF-2	150 cfm	155 cfm	Sample
EF-3	150 cfm	140 cfm	Sample

Page 2 is a landscape page stored with a rotation, as scanners often save them.

BALANCED AIR TESTING (FICTIONAL)

Harborview Family Clinic (fictional) · Project 2026-014

Diffuser readings (sample values)

Room	Diffuser	Design cfm	Measured cfm	Percent	Notes
Exam 1	S-1	180	173	95%	Sample
Exam 2	S-2	180	176	96%	Sample
Exam 3	S-3	180	179	97%	Sample
Exam 4	S-4	180	182	98%	Sample
Waiting	S-5	250	244	98%	Sample
Reception	S-6	150	151	101%	Sample

Final test and balance report

7.2 · Section 7 — Testing, Adjusting and Balancing (23 05 93)

This document has not been received yet. Replace this page with the document when it arrives.

Note: To be provided by the balancing contractor

Requested on: _____ Received on: _____

SECTION

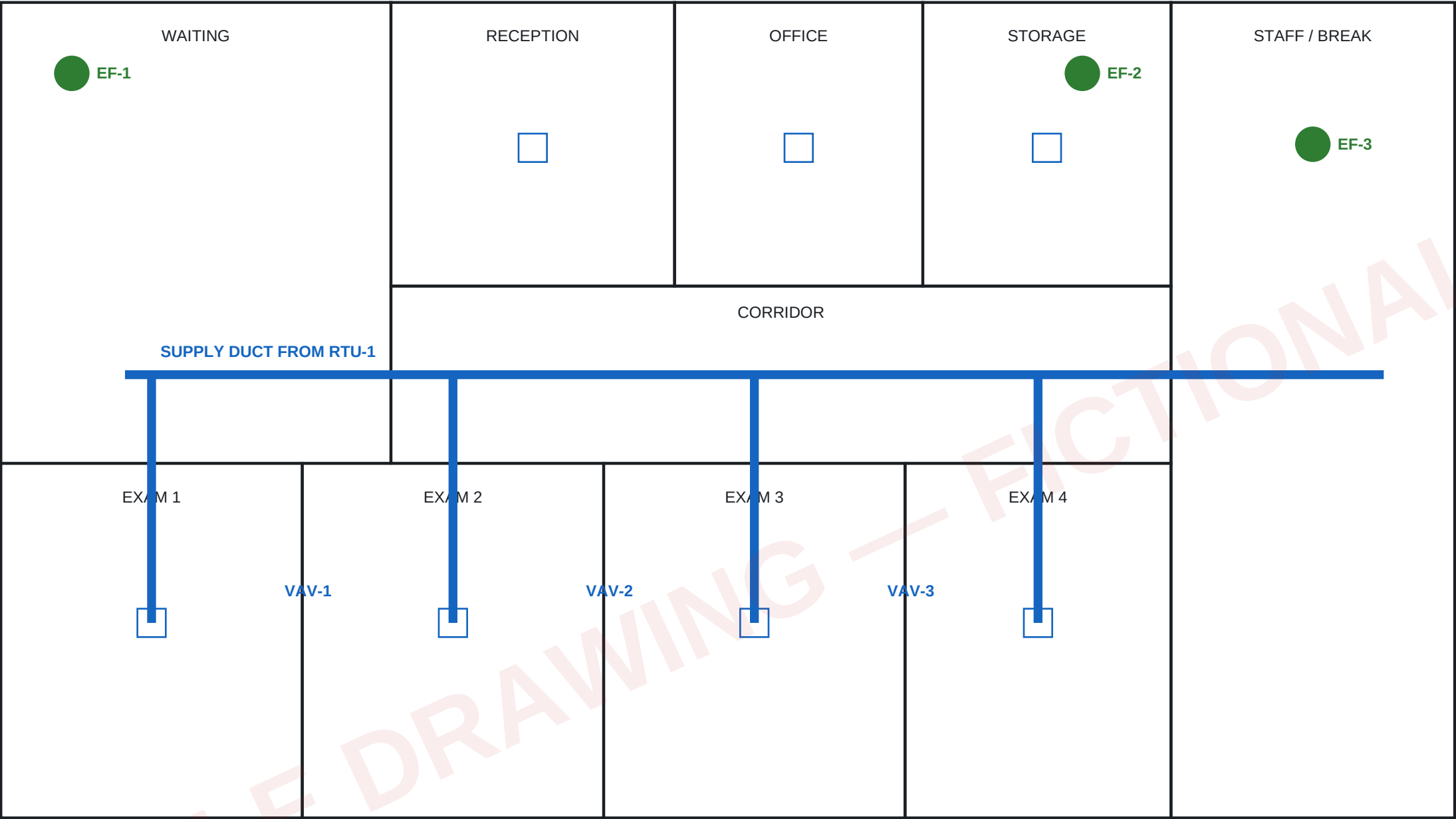
8

Record Drawings

IN THIS SECTION

PAGE

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Sample drawing for a portfolio binder. Not to scale. Not for construction.

HARBORVIEW FAMILY CLINIC

Tenant fit-out (fictional project) · 2026-014
MECHANICAL PLAN - HVAC

M-101

RECORD DRAWING (SAMPLE)